Height of the window = h metres

$= AQ\ (WR = AQ)$

Let $PA = x$ metres.

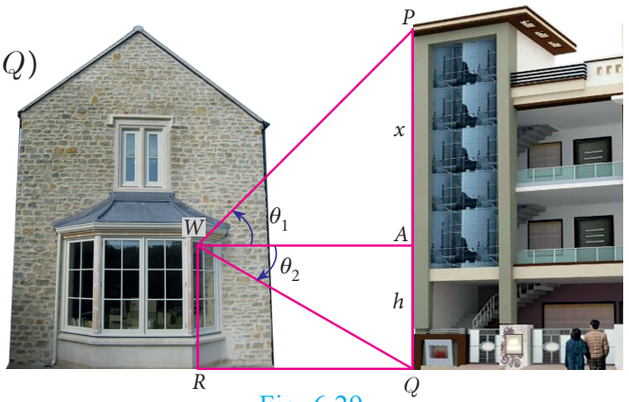


Fig. 6.29

In the right angled ΔPAW, $\tan\theta_1 = \dfrac{AP}{AW}$

$\Rightarrow \quad \tan\theta_1 = \dfrac{x}{AW}$

$AW = \dfrac{x}{\tan\theta_1}$

we get, $AW = x\cot\theta_1$ ...(1)

In the right angled ΔQAW, $\tan\theta_2 = \dfrac{AQ}{AW}$

$\Rightarrow \quad \tan\theta_2 = \dfrac{h}{AW}$

we get, $AW = h\cot\theta_2$ ...(2)

From (1) and (2) we get, $x\cot\theta_1 = h\cot\theta_2$

$\Rightarrow \quad x = h\dfrac{\cot\theta_2}{\cot\theta_1}$

Thinking Corner

What is the minimum number of measurements required to determine the height or distance or angle of elevation?

Therefore, height of the opposite house = $PA + AQ = x + h = h\dfrac{\cot\theta_2}{\cot\theta_1} + h = h\left(1 + \dfrac{\cot\theta_2}{\cot\theta_1}\right)$

Hence Proved.

Progress Check

1. The line drawn from the eye of an observer to the point of object is _____.
2. Which instrument is used in measuring the angle between an object and the eye of the observer?
3. When the line of sight is above the horizontal level, the angle formed is _____.
4. The angle of elevation _____ as we move towards the foot of the vertical object (tower).
5. When the line of sight is below the horizontal level, the angle formed is _____.

Exercise 6.4

1. From the top of a tree of height 13 m the angle of elevation and depression of the top and bottom of another tree are $45°$ and $30°$ respectively. Find the height of the second tree. $(\sqrt{3} = 1.732)$

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